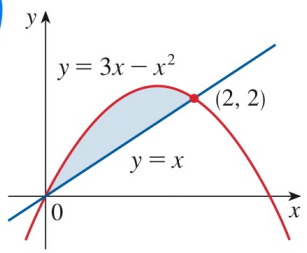
Exercises 6.1 1, 3, 7, 13, 15, 17, 21, 25, 27, 57, 61.

1–4

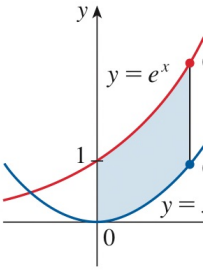
(a) Set up an integral for the area of the shaded region.

(b) Evaluate the integral to find the area.

1.



2.



1. $A = \int_0^2 3x - x^2 - x \, dx$

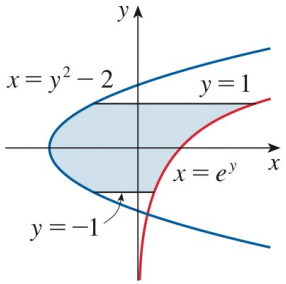
$= \int_0^2 2x - x^2 \, dx$

$= \left[x^2 - \frac{x^3}{3}\right]_0^2$

$= \left(2^2 - \frac{2^3}{3}\right) = \frac{4}{3} = 1.333$

$e - y^2 + 2$

3.



$A = \int_{-1}^{1} e^y - (y^2 - 2) \, dy$

$= \int_{-1}^{1} e^y - y^2 + 2 \, dy$

$= \left[e^y - \frac{y^3}{3} + 2y\right]_{-1}^{1}$

$= \left(e - \frac{1}{3} + 2\right) - \left(e^{-1} + \frac{1}{3} - 2\right)$

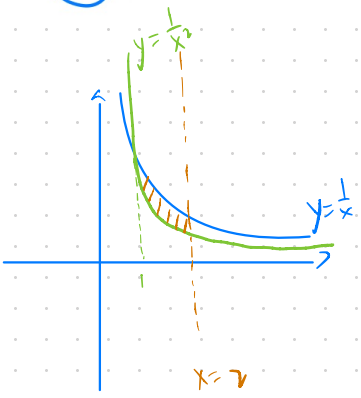
$= 5.6837$

11–18 Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle and label its height and width. Then find the area of the region.

11. $y = x^2 + 2, \quad y = -x - 1, \quad x = 0, \quad x = 1$

12. $y = 1 + x^3, \quad y = 2 - x, \quad x = -1, \quad x = 0$

13. $y = 1/x, \quad y = 1/x^2, \quad x = 2$



$\frac{1}{x} = \frac{1}{x^2}$

$x = 1$

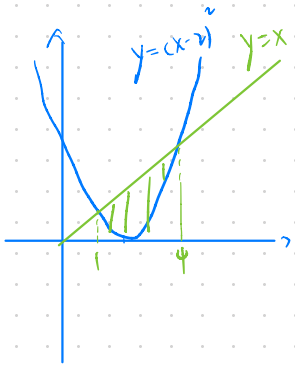
$A = \int_1^2 \frac{1}{x} - \frac{1}{x^2}\, dx$

$= [\ln|x| + x^{-1}]_1^2$

$= \ln|2| + \frac{1}{2} - (\ln|1| - 1)$

$= 2.193$

15. $y = (x - 2)^2, \quad y = x$



$(x-2)^2 = x$

$x^2 - 4x + 4 - x = 0$

$x^2 - 5x + 4 = 0$

$x = 1$ or $x = 4$

$x - (x^2 - 4x + 4)$

$\int_1^4 x - (x-2)^2\, dx$

$= \int_1^4 x - x^2 + 4x - 4\, dx$

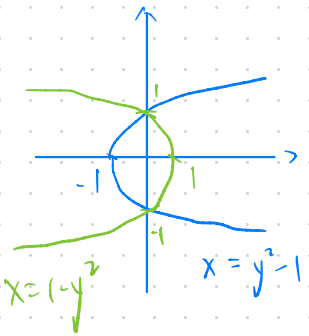
$= \int_1^4 -x^2 + 5x - 4\, dx$

$= \left[-\frac{1}{3}x^3 + \frac{5}{2}x^2 - 4x\right]_1^4 = \left(-\frac{1}{3}\cdot 4^3 + \frac{5}{2}\times 4^2 - 4\times 4\right) - \left(-\frac{1}{3} + \frac{5}{2} - 4\right)$

$= \frac{8}{3} + \frac{11}{6}$

$= \frac{9}{2} = 4.5$

17. $x = 1 - y^2, \quad x = y^2 - 1$



$$A = \int_{-1}^{1} 1 - y^2 - y^2 + 1 \, dy$$

$$= 2\int_{-1}^{1} 1 - y^2 \, dy$$

$$= 2\left[y - \frac{y^3}{3}\right]_{-1}^{1}$$

$$= 2\left[\left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right)\right]$$

$$= \frac{8}{3}$$